

# JEE MAIN REVISION MASTER: PERMUTATIONS AND COMBINATIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 14 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                                 | Frequency | Category                  |
|---------------|--------------------------------------------------------------|-----------|---------------------------|
| High          | Fundamental Principles & Permutations (Distinct & Identical) | 11 PYQs   | Arrangements              |
| Moderate      | Combinations, Selection & Multinomial Partitioning           | 4 PYQs    | Selections & Distribution |

## 2. Master Formula Reference Card

| Concept / Topic                                              | Mathematical Formulation & Governing Equations                                                                                                            |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fundamental Principles & Permutations (Distinct & Identical) | ${}^n P_r = \frac{n!}{(n-r)!}$ , $\text{Arrangements with duplicates: } \frac{n!}{p! q! r!}$ , $\text{Circular: } (n-1)!$                                 |
| Combinations, Selection & Multinomial Partitioning           | ${}^n C_r = \frac{n!}{r!(n-r)!}$ , $\text{Non-negative solutions: } {}^{n+r-1} C_{r-1}$ , $\text{Derangements: } D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!}$ |

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Fundamental Principles & Permutations (Distinct & Identical) (High Yield — 11 PYQs)

**Core Principle:** Multiplication and addition principles, linear permutations with repetitions, rank of words in dictionary, and circular arrangements.

■ **Caution / JEE Pitfall:** For circular permutations of necklaces/garlands where clockwise and anticlockwise arrangements are indistinguishable, divide by 2:  $((n-1)! / 2)$ .

■ **Speed Trick:** Gap method: Place items that cannot be adjacent into the gaps created by the remaining unrestricted items.

**Representative Source PYQs:**

- [2026 Shift] Q#1 (MCQ): "How many ways can you arrange all the letters and numbers in "KCET 2025" which start with K and end with 5 ?..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "The number of four digit even number that can be formed using the digits 0,1,2 and 3 without repetition is..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "The number of diagonals that can be drawn in an octagon is..." [Ans: N/A]

### 3.2 Combinations, Selection & Multinomial Partitioning (Moderate Yield — 4 PYQs)

**Core Principle:** Combinatorial selection, Pascal identity  ${}^nC_r + {}^nC_{(r-1)} = {}^{n+1}C_r$ , distribution of identical items into distinct bins, and derangements.

■ **Caution / JEE Pitfall:** Number of positive integral solutions ( $x_i \geq 1$ ) to  $x_1 + x_2 + \dots + x_r = n$  is  ${}^{n-1}C_{(r-1)}$ , whereas non-negative ( $x_i \geq 0$ ) is  ${}^{n+r-1}C_{(r-1)}$ .

■ **Speed Trick:** Number of derangements of 4 items is  $D_4 = 9$ ; for 5 items is  $D_5 = 44$ .

**Representative Source PYQs:**

- **[2026 Shift] Q#2 (MCQ):** "10 distinct points are taken on a circle. Then using these points Statement I: The number of triangles that can be formed is 100 Statement II: The number of chords that can be formed is 45 Which of the following is correct?" **[Ans: N/A]**
- **[2023 Shift] Q#1 (MCQ):** "Ten chairs are numbered as 1 to 10. Three women and two men wish to occupy one chair each. First the women choose the chairs marked 1 to 6, ..." **[Ans: A]**
- **[2018 Shift] Q#4 (MCQ):** "Everybody in a room shakes hands with everybody else. The total number of handshakes is 45. The total number of persons in the room is..." **[Ans: N/A]**

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: KCET 2026 (Q#1)

How many ways can you arrange all the letters and numbers in "KCET 2025" which start with K and end with 5 ?

- (A) 720
- (B) 360
- (C) 120
- (D) 180

**Correct Answer: None**

**Solution Analysis:** The characters in "KCET 2025" are: K, C, E, T, 2, 0, 2, 5. There are 8 characters in total, and the digit 2 is repeated twice. We need arrangements that start with K and end with 5. So fix K in the first position and 5 in the last position: K\_\_\_\_\_5. Now we have to arrange the remaining 6 characters: C, E, T, 2, 0, 2. If all were different, the number of arrangements would be:  $6!$ . But the digit 2 appears twice, so we divide by  $2!$ :  $(6! / 2!) = (720 / 2) = 360$ . So the number of ways is: 360. Correct option: Option B.

### Benchmark Problem 2: KCET 2026 (Q#2)

10 distinct points are taken on a circle. Then using these points Statement I: The number of triangles that can be formed is 100 Statement II: The number of chords that can be formed is 45 Which of the following is correct?

- (A) Both Statement I and Statement II are true
- (B) Both Statement I and Statement II are false
- (C) Statement I is true and Statement II is false
- (D) Statement I is false and Statement II is true

**Correct Answer: None**

**Solution Analysis:** For a triangle, we need to choose 3 points from the 10 distinct points on the circle. Since all points lie on a circle, no three points are collinear. So every selection of 3 points forms a triangle. Number of triangles:  ${}^{10}C_3 = (10 \times 9 \times 8 / 3 \times 2 \times 1) = 120$ . So Statement I is false, because the number of triangles is not 100. For a chord, we need to choose 2 points on the circle. Every pair of points gives one chord. Number of chords:  ${}^{10}C_2 = (10 \times 9 / 2 \times 1) = 45$ . So Statement II is true. Hence, the correct option is: Option D.